



Introduction to ECG Interpretation

Understanding 12-Leads, ECG Components, Artifacts & Lead Morphology

COMPILED AND PRESENTED

BY

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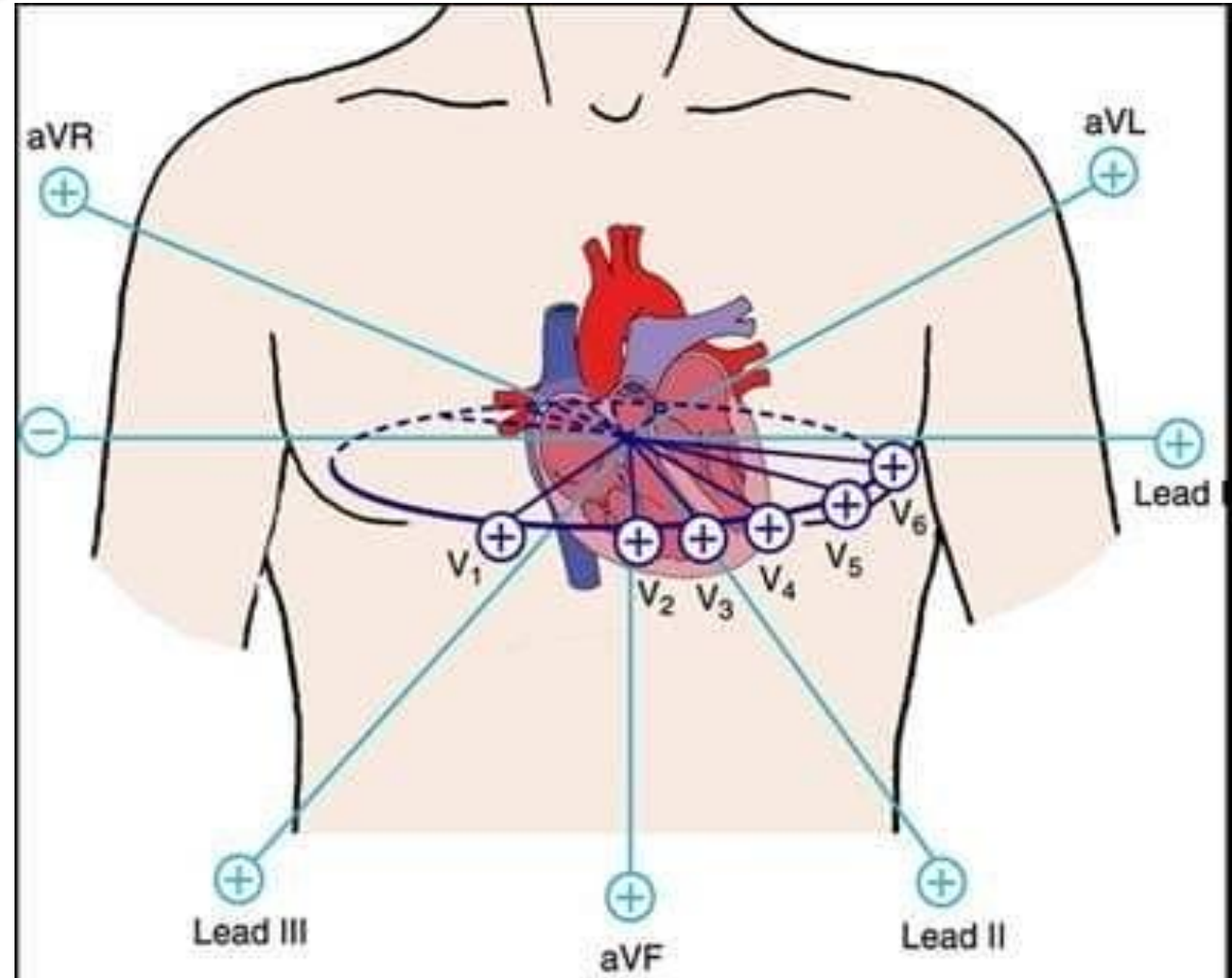
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Learning Objectives

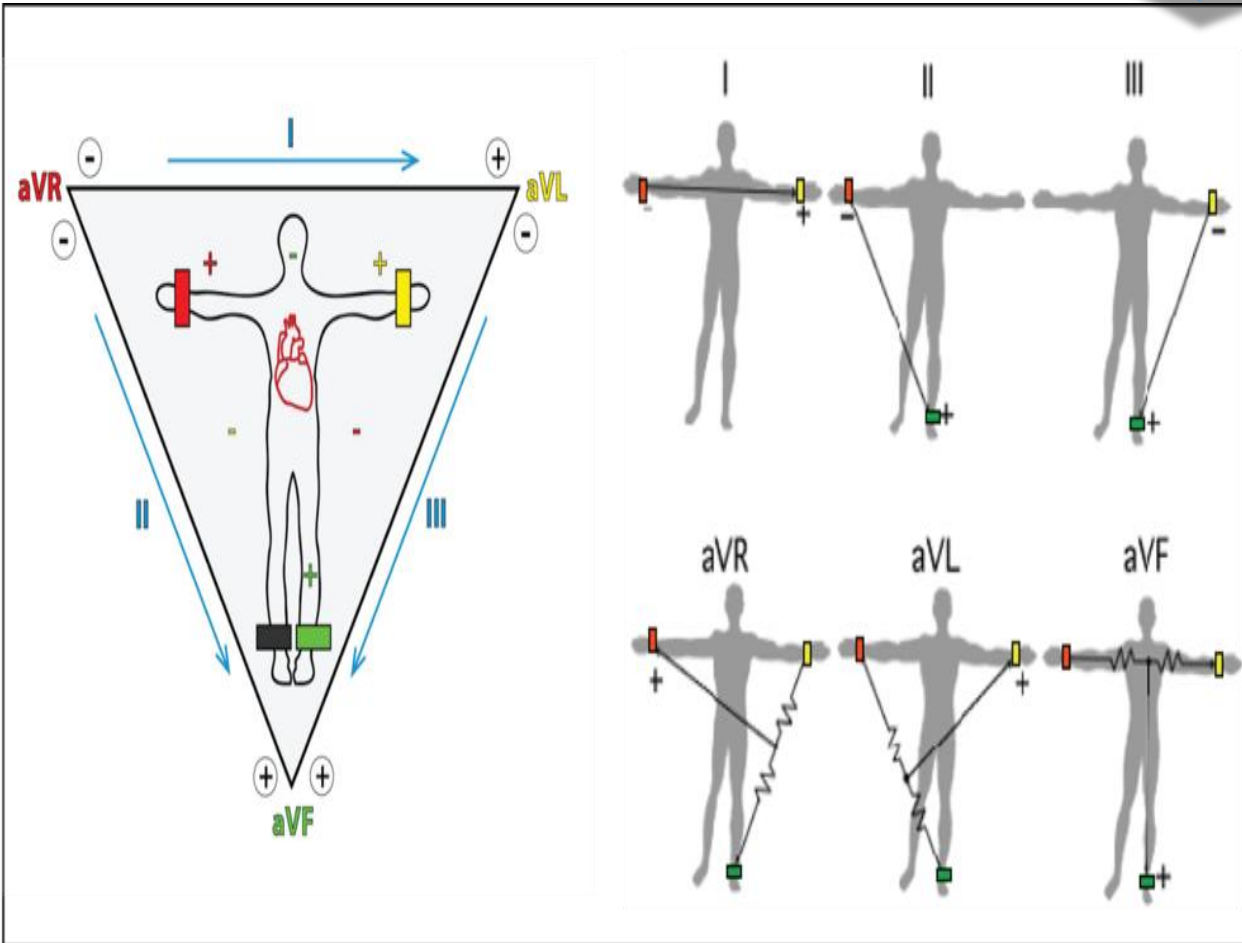
- Understand the concept of the 12-lead ECG and the Einthoven Triangle.
- Recognize the key waves, segments, and intervals of a standard ECG.
- Identify common ECG artifacts and their causes.
- Understand normal lead morphology and what it tells us about the heart.

What is a 12-Lead ECG?

- Definition: A recording of the heart's electrical activity from 12 different perspectives.
- Includes 3 limb leads, 3 augmented limb leads, and 6 chest (precordial) leads.



The Einthoven Triangle



- Named after Willem Einthoven, who invented the ECG.
- Formed by leads I, II, and III across the limbs (RA, LA, LL).
- **Lead I:** LA (+) to RA (-)
- **Lead II:** LL (+) to RA (-)
- **Lead III:** LL (+) to LA (-)

Augmented Limb Leads & Chest Leads

- **aVR**: RA to heart center (the lead attached to right arm (- deflection)).
- **aVL**: LA to heart center (the lead attached to left arm (+ deflection)).
- **aVF**: LL to heart center (the lead attached to the left foot (+ deflection)).

a = augmented; V = voltage

Chest leads: V1–V6: Chest leads placed from 4th ICS to midaxillary line on the 5th ICS.

The ECG Complex Overview

- ECG records electrical depolarization and repolarization.
- Key components: **P wave**, **QRS complex**, **T wave**
- Depolarization – electrical activation → muscle contraction.
- Repolarization – return to resting state → muscle relaxation.
- Systole = Contraction, Diastole = Relaxation

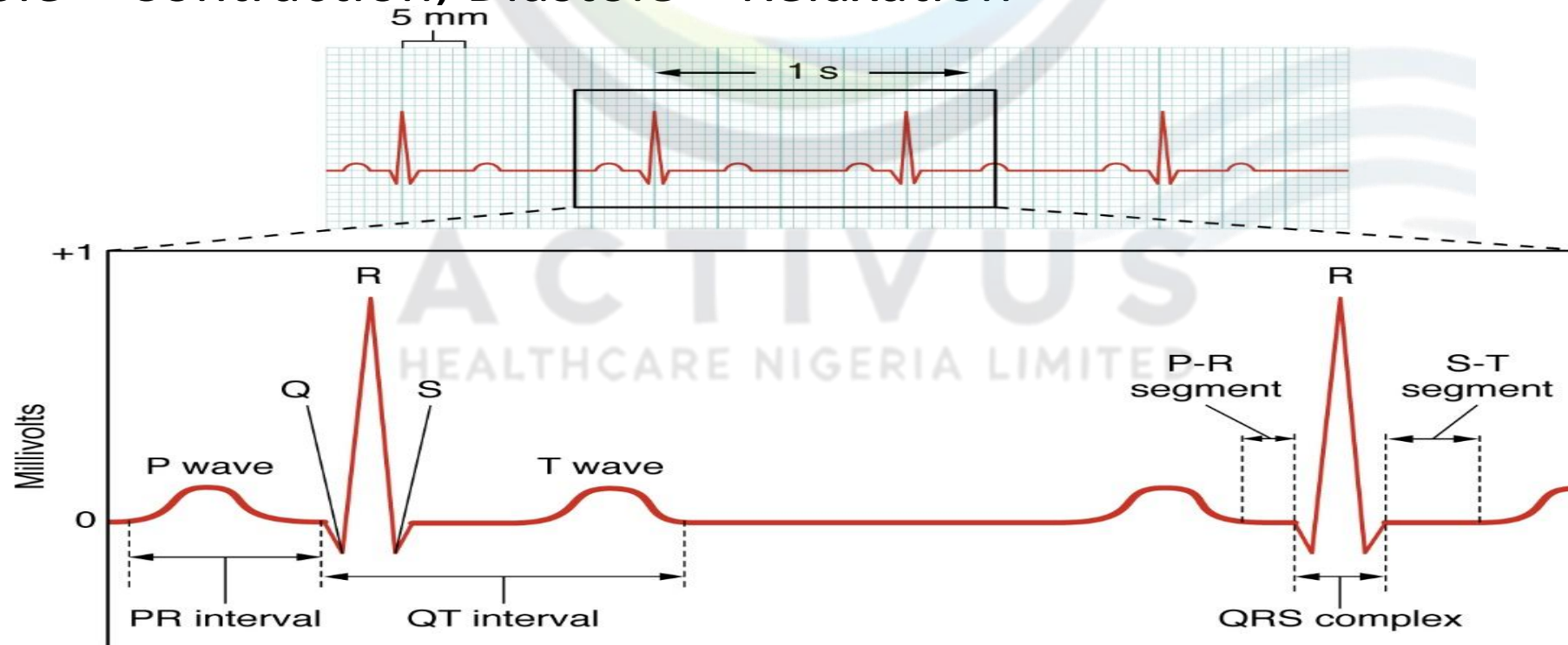
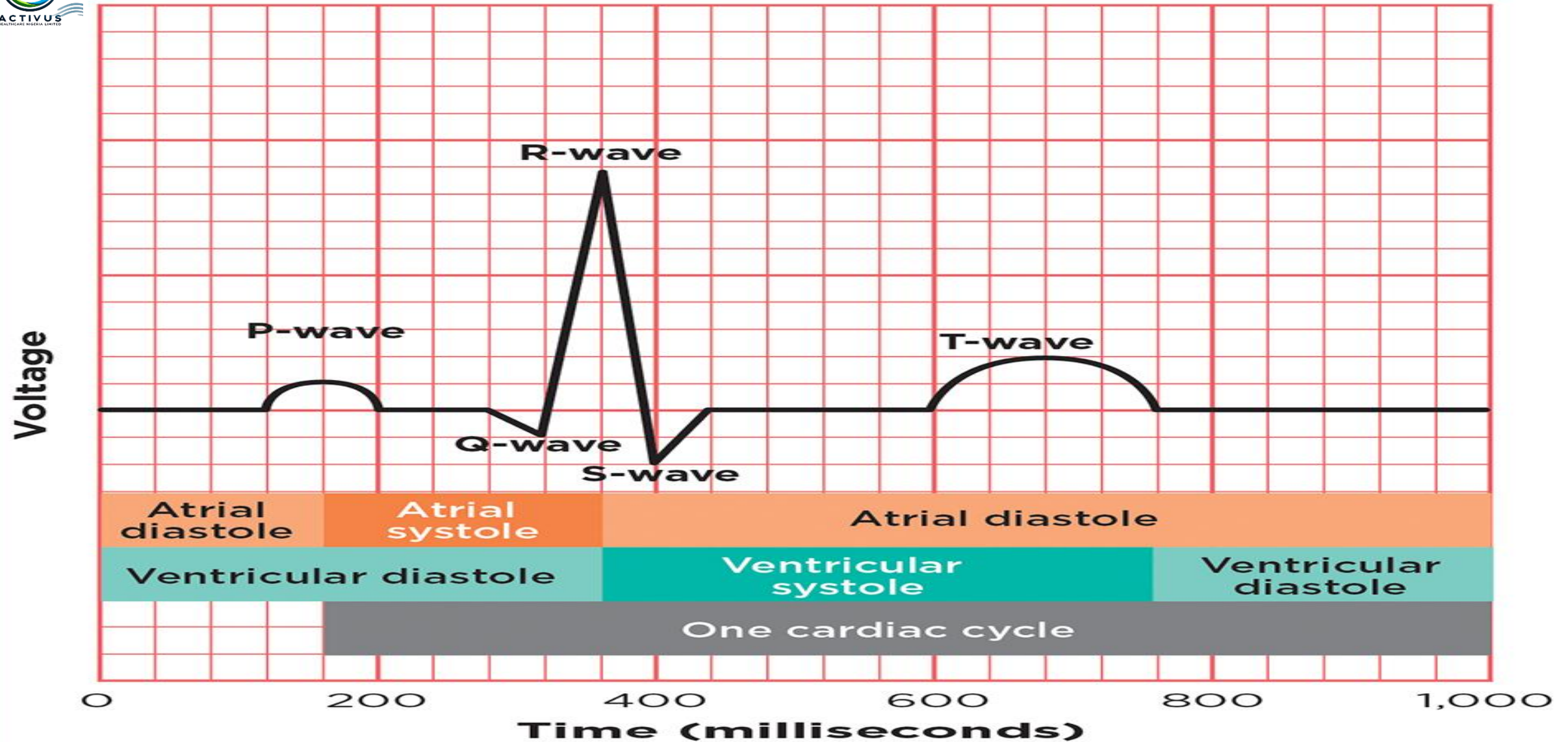
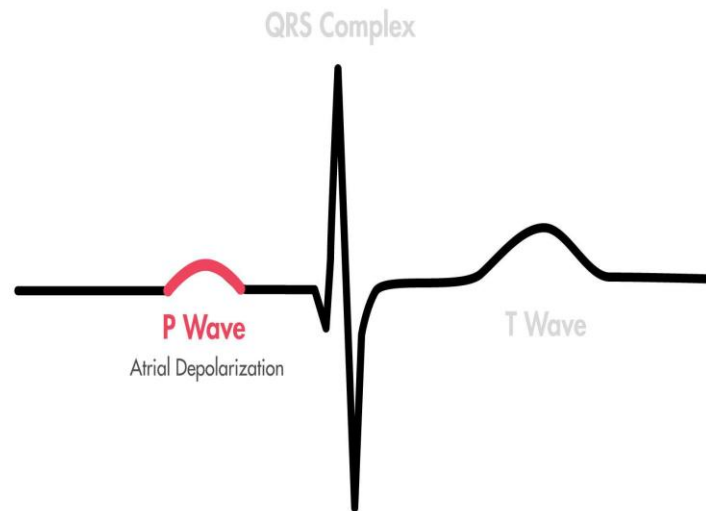


Fig 2. Relationship between ECG and cardiac cycle stages



P Wave

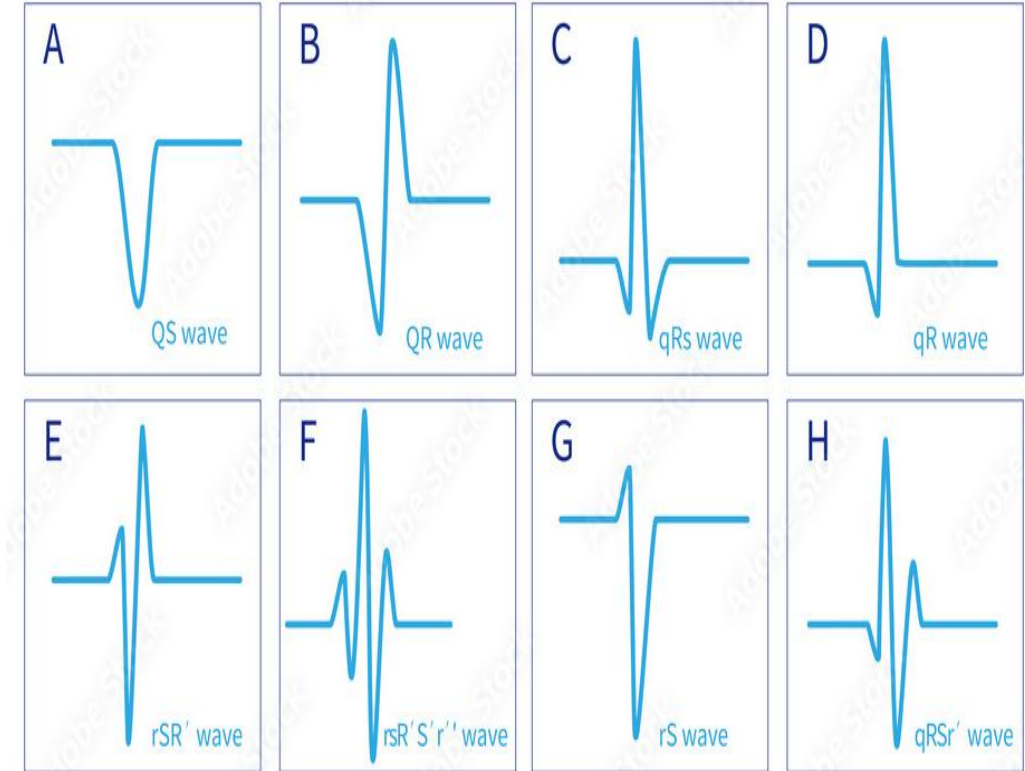
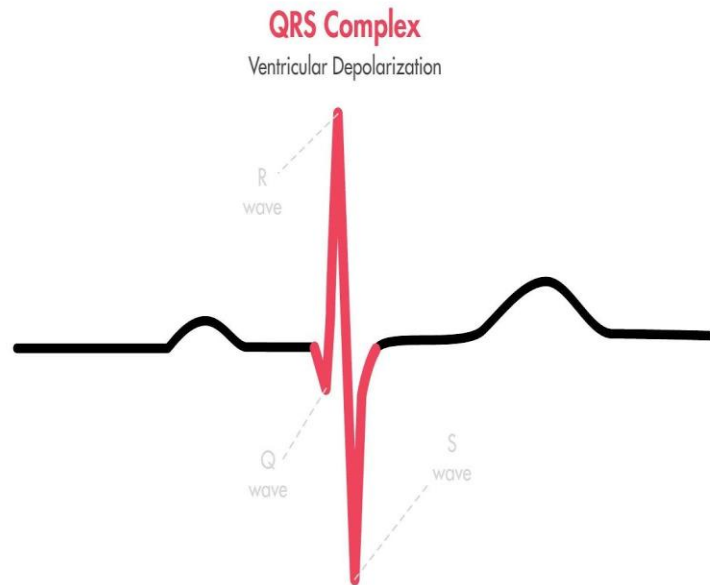
- Atrial depolarization
- Duration: $< 0.12s$, Amplitude: $< 2.5mm$ in limb leads
- **Clinical Insight:** Peaked P may indicate right atrial enlargement.



	II	V1
Normal		
RAE		
LAE		
RAE + LAE		

QRS Complex

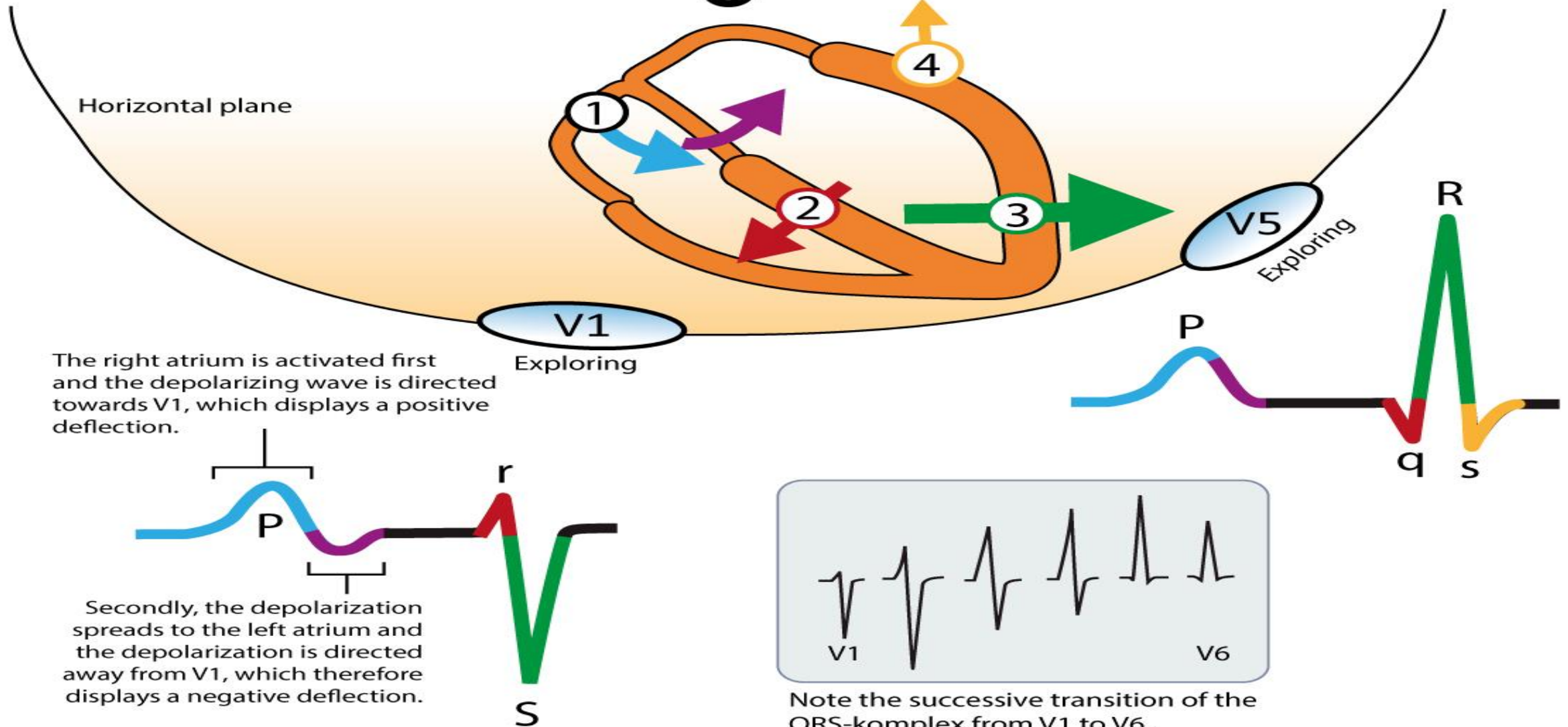
- Ventricular depolarization
- Duration: $< 0.12s$
- Q wave: first negative deflection
- R wave: first positive deflection
- S wave: negative deflection after R
- QRS wave amplitude greater than 5 mm or 0.5 mV is written in capital Letter e.g. Q, R and S.
- QRS wave amplitude lesser than 5 mm or 0.5 mV is written in small letter e.g. q, r and s.



Naming of QRS complexes

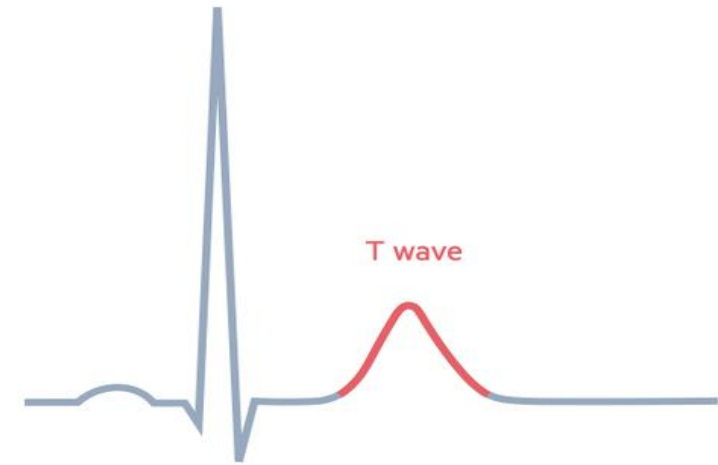
The reference point (Wilson's central terminal) is constructed by taking the average of all three limb electrodes. Taking the average of these yields a reference point located in the centre of Einthoven's triangle.

Reference 



T Wave and U wave

- T wave: Ventricular repolarization
- U wave (if present): U wave (+) is produced by depolarisation of the papillary muscles – an inconstant finding or electrolyte disturbance
- **Clinical Insight:** Inverted T waves may suggest ischemia.



Inverted



Peaked



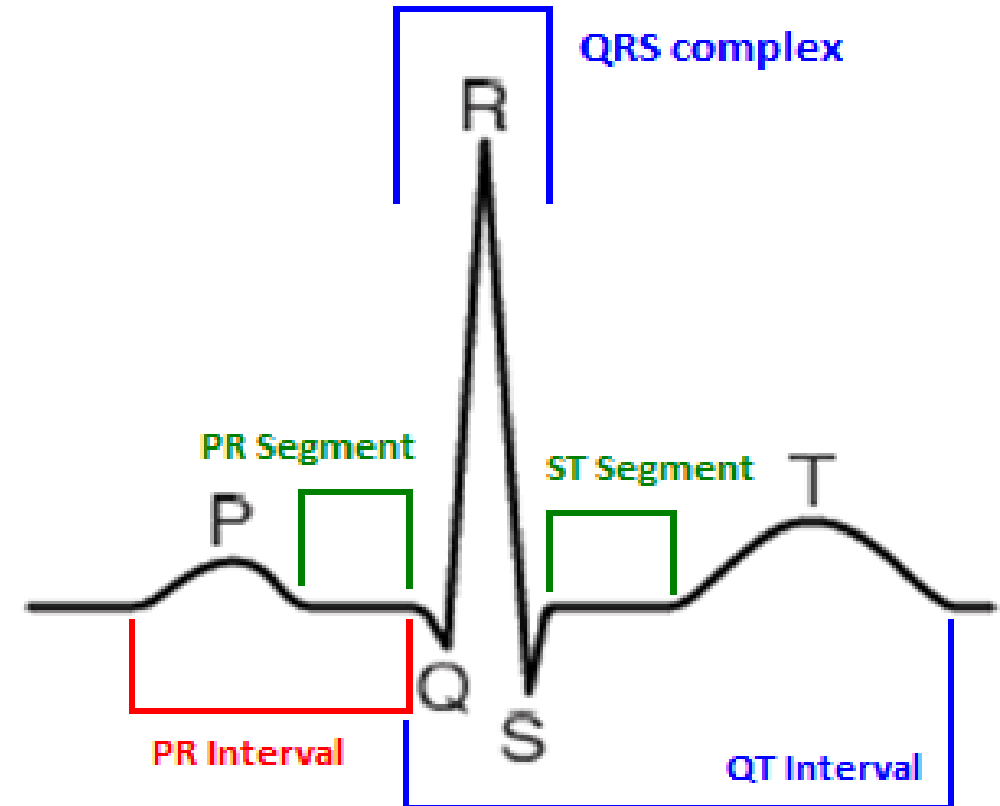
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Bifid

Intervals and Segments

- **PR Interval:** Atrial to ventricular conduction (0.12–0.20s).
- **QT Interval:** (about 0.40s) Total time for ventricular depolarization and repolarization (corrected = QTc)
- **ST Segment:** Time between depolarization and repolarization.



What are Artifacts?

- Unwanted signals that can mimic or obscure cardiac activity.
- Can lead to misinterpretation if not recognized.

Common Types of Artifacts

- **Muscle Tremor:** Irregular baseline from muscle activity.
- **Motion Artifact:** Large spikes from patient movement.
- **60-cycle Interference:** Uniform small waves from electrical interference.
- **Loose Electrode:** Baseline wander or dropped signals.



TYPES OF ARTIFACT



LOOSE LEAD



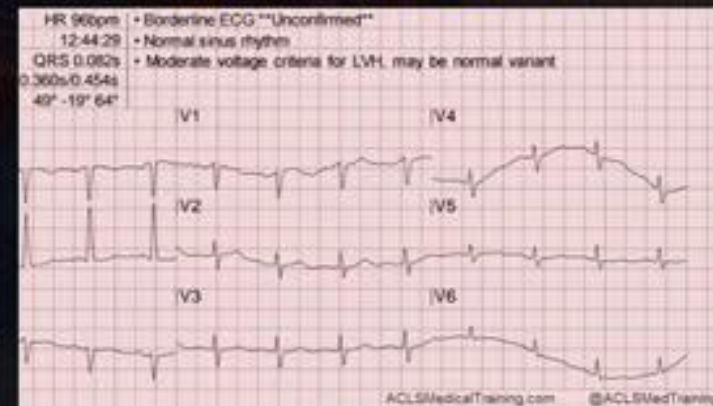
- ✓ Check lead connection
- ✓ Replace electrode

60 CYCLE INTERFERENCE



- ✓ Check that the RL electrode is attached
- ✓ If monitor allows, filter 0.05hz - 40hz

WANDERING BASELINE



- ✓ Commonly caused by patient movement

MUSCLE TREMOR



- ✓ Commonly caused by shivering/tremors

NEUROMODULATION



- ✓ Ask patient if they have any implanted devices (neuro stimulators)

ARTERIAL PULSE TAPPING



- ✓ Make sure the electrodes are not placed over top of a bounding artery or fistula

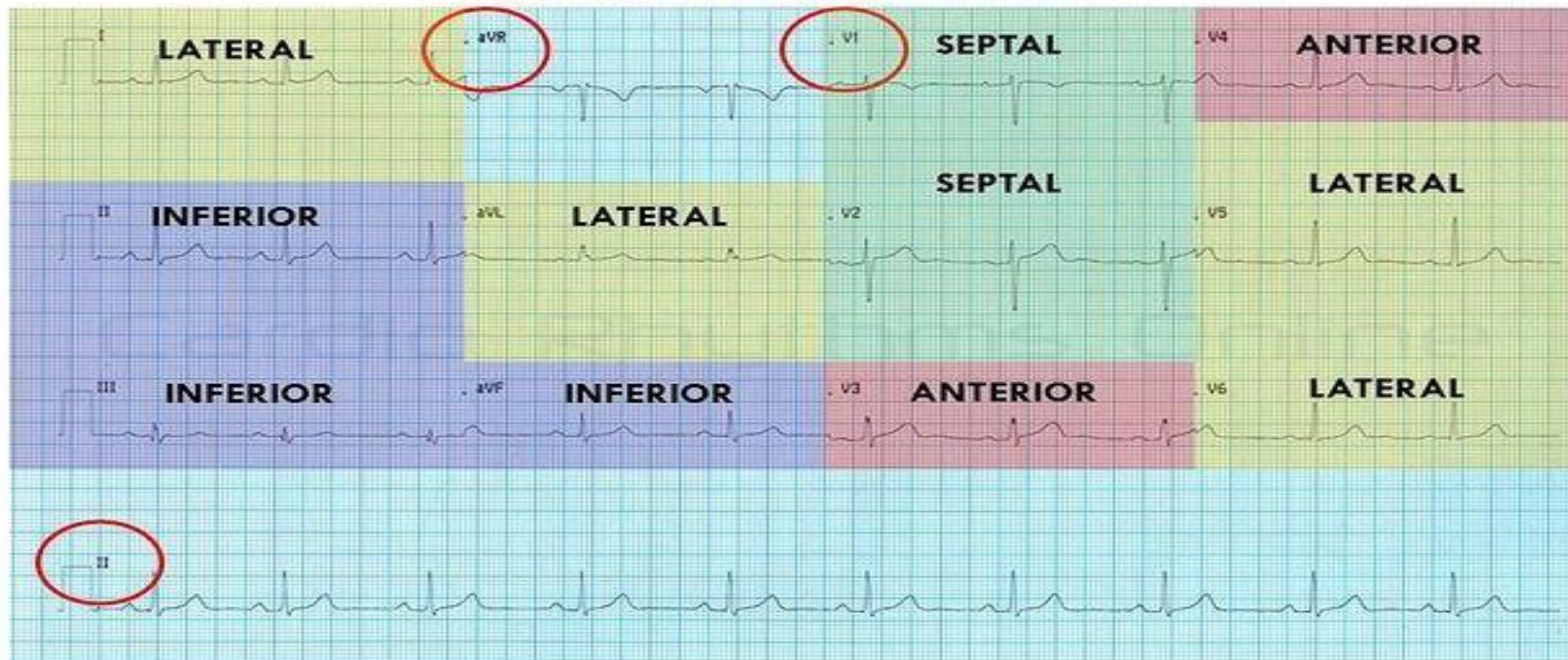
Reducing Artifacts

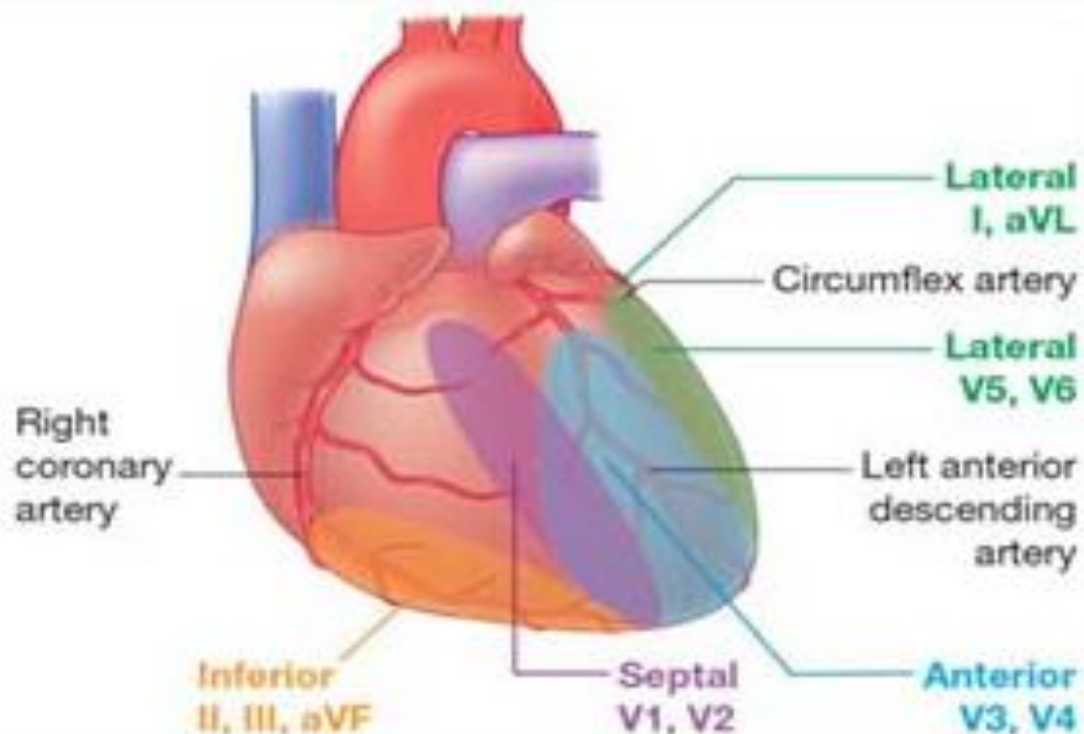
- Ensure good electrode-skin contact.
- Patient still and relaxed.
- Avoid electrical devices near ECG.
- Use alcohol wipes before placing electrodes.

Understanding Lead Morphology

- Different leads view the heart from different angles
- Each lead has a characteristic waveform shape in a healthy heart

The 12-Lead ECG Explained





The inferior leads look at the inferior surface (or underneath) of the heart. Usual blood supply: right coronary artery

The septal leads look at the interventricular septum (LAD)

The lateral leads look at the lateral (or side) walls of the left ventricle. Usual blood supply: circumflex branch of left coronary artery

The anterior leads look at the anterior surface (or front wall) of the right and left ventricle. Usual blood supply: left anterior descending (LAD) branch of left coronary artery

Description	ECG leads with changes	Artery occluded
Inferior	II, III, aVF	RCA
Anteroapical	V3 and V4	Distal LAD
Anteroseptal	V1 and V2	LAD
Anterolateral	I, aVL, V5 and V6	Circumflex artery
Extensive anterior	I, aVL, V2-V6	Proximal LCA
True posterior	Tall R in V1	RCA

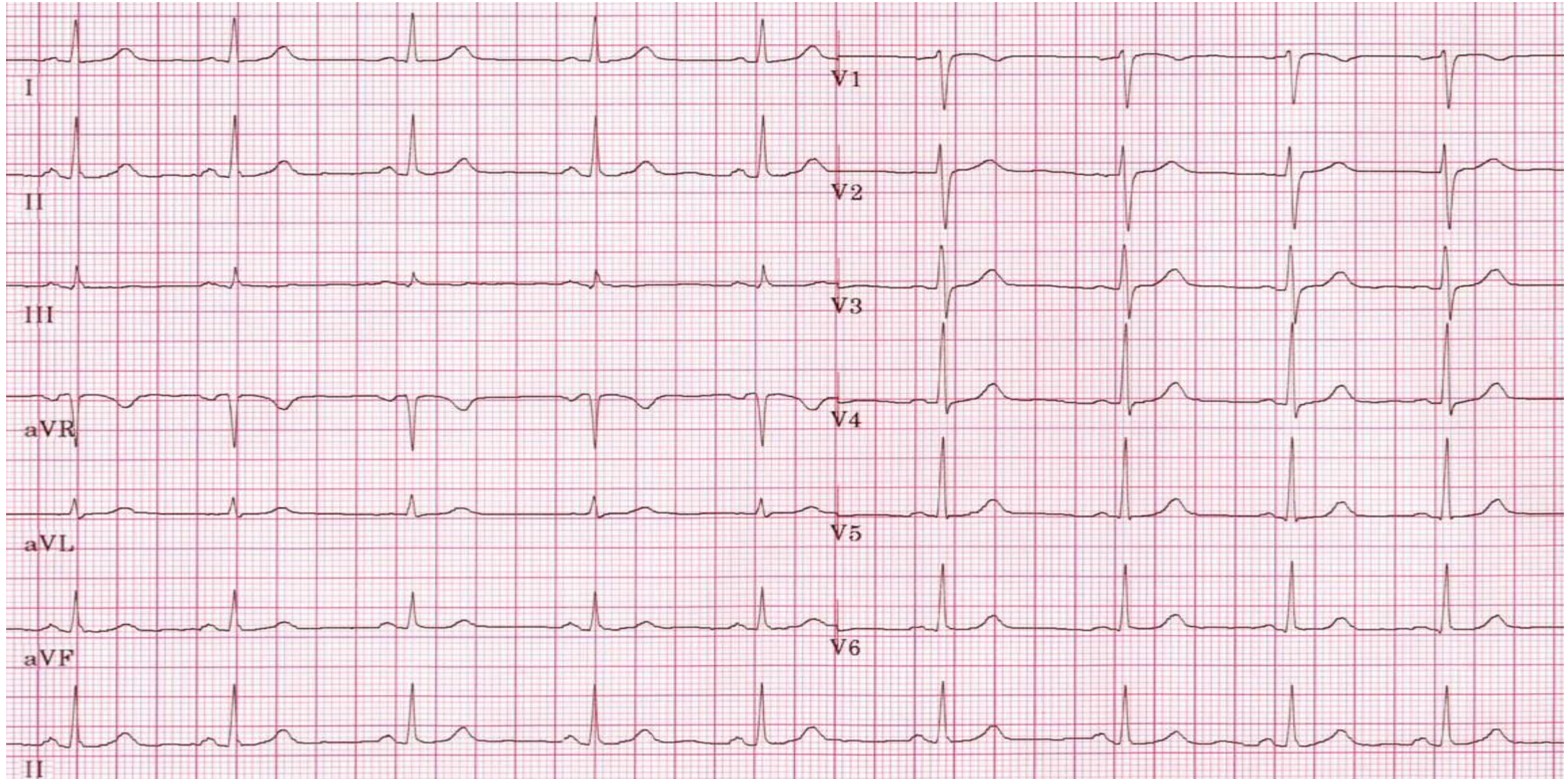
Limb Lead Morphology

- Lead I: Upright P, QRS, T
- Lead II: Ideal for rhythm – upright P, QRS, T
- aVR: Typically inverted waves
- aVL: Small amplitude; QRS can be biphasic or upright.
- aVF: Upright P, QRS, and T waves.

Precordial Lead Progression

- V1–V2: Mostly negative QRS
- V3–V4: Transition zone
- V5–V6: Mostly positive QRS
- **Clinical Insight:** Poor R-wave progression may suggest anterior MI.

Normal 12 lead ECG strip



Summary

- 12-lead ECG gives a complete electrical picture of the heart.
- Key components: P wave, QRS complex, T wave, PR/QT intervals.
- Recognizing artifacts is essential for accurate diagnosis.
- Lead morphology varies and helps in locating abnormalities.

Question

Why is lead II considered best for rhythm strip and determination of sinus rhythm?

References & Resources

- Books: *Rapid Interpretation of ECGs* by Dale Dubin, *ECG Made Easy* by John Hampton.
- Online simulators and ECG practice platforms (e.g., ECGpedia, Life in the Fast Lane, Google).

